Fusion of Multisource Satellite AOD Products via Bayesian Maximum Entropy With Explicit A Priori Knowledge

Hao Zhu, Tianhai Cheng, Xiaoyang Li, Xiaotong Ye, Donghao Fan, Tao Tang, and Lili Zhang

***Abstract*— Fusing multiple satellite aerosol optical depth (AOD) products to produce high-quality aerosol records is necessary for climate-related research. The Bayesian maximum entropy (BME) approach has unique advantages in this regard. However, the large accuracy differences and redundancy characteristics between aerosol products are not considered by BME resulting in limited fusion quality and large computational consumption. Therefore, in this study, we try to explicitly introduce a priori knowledge about the multisource AOD product accuracy, called the accuracy ranking matrix, into the BME fusion process to test whether the above issues can be alleviated. Eight publicly released aerosol products over China are used in the fusion experiments, and the fusion results are validated with ground-based measurements. Results suggest that, compared with fusion directly, the fusion accuracy and efficiency are significantly improved by introducing the accuracy ranking matrix; the proportion of satisfying the Global Climate Observing System (GCOS) accuracy requirements increases by 12.2%, and the consumption time decreases by 74.6%. Specifically, the accuracy improvement in vegetated and clean air conditions outperforms that in arid and polluted conditions, with the most significant improvement in clean conditions, where the GCOS fraction increased by 22.9%. Moreover, the spatial distribution of fusion results in typical regions indicates that the introduction of the accuracy ranking matrix makes the AOD spatial variation smoother. These results demonstrate that introducing the accuracy ranking matrix can generate higher quality AOD distributions with improved efficiency, which is expected to guide the fusion of AOD products or other remote sensing products.**

***Index Terms*— Accuracy ranking matrix, Bayesian maximum entropy (BME), multiple product fusion, satellite aerosol optical depth (AOD) products.**

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I. Introduction

ATMOSPHERIC aerosols affect the climate system directly by scattering and absorbing solar radiation [1], [2] and indirectly by changing cloud properties [3], [4], which is the main uncertainty source in climate-related studies [5]. Aerosol optical depth (AOD), which describes light extinction by atmospheric aerosols [6], is an important optical parameter for characterizing aerosol particles and understanding their role in the environment and climate [7]. Due to their global coverage capability, satellites can provide large-scale AOD distribution information, making them an important tool for investigating the climate impacts of aerosols [8], [9], [10].

Currently, an increasing number of satellite sensors have released aerosol products based on various retrieval algorithms, such as the Moderate Resolution Imaging Spectroradiometer (MODIS) onboard the Terra/Aqua [11], [12], the Multiangle Imaging SpectroRadiometer (MISR) onboard the Terra [13], the Ocean and Land Color Instrument (OLCI) and Sea and Land Surface Temperature Radiometer (SLSTR) onboard the Sentinel-3 [14], and the Advanced Himawari Imager (AHI) onboard the Himawari-8 [15]. However, the spatial completeness of these aerosol products is limited due to cloud contamination and assumptions in the retrieval algorithm [16], [17]. Moreover, differences in sensor sampling and retrieval algorithms lead to inconsistencies in accuracy [18], [19], spatial distribution [20], [21], and temporal trends [22], [23] among various aerosol products, which may cause large uncertainties in climate-related studies [24], [25]. In our previous study, we compared the accuracy of 16 currently publicly released satellite aerosol products in different scenarios and demonstrated that they each have their own strengths and weaknesses, and no one product performs optimally in all scenarios [18]. Therefore, the complementary characteristics of these AOD products make it possible to combine their strengths to produce AOD distribution data with more complete spatial coverage and high quality to meet the needs of climate research users [26].

Due to rapid industrialization and urbanization, China, especially the Beijing–Tianjin–Hebei (BTH) region and the Yangtze River Delta (YRD) region, has severe atmospheric aerosol loading and complex aerosol components [27], [28]. Spatially complete, consistent, and high-quality AOD data are

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needed to better understand the environmental and climate effects of aerosols in order to address the corresponding scientific questions and establish environmental policies. Previous studies have developed various algorithms to fuse different satellite AOD products, which can be divided into single-pixel-based methods and multipixel-based methods. Single-pixel-based methods provide fused AOD values only at pixels where the original satellite product has a valid AOD value, and this value is obtained by a functional relationship between different products. Such methods include linear or polynomial functions [26], [29], optimal interpolation [30], [31], and maximum likelihood methods [32]. However, this approach fails in areas where none of the original satellite AOD products have valid values, resulting in fusion products that are still limited in spatial completeness. In contrast, multipixel-based methods take into account the effect of adjacent pixels based on correlation, so that the AOD of missing pixels can also be estimated, such as the geostatistical inversion model [33], spatial statistics fusion (SSDF) method [34], the fixed rank smooth method [35], universal kriging method [36], and the ensemble Kalman filter method [37]. In addition to spatial autocorrelation, temporal autocorrelation of AOD is also the necessary information to effectively enhance the completeness of the fused AOD, especially in the areas where the original satellite AOD is missing significantly [38], [39]. Consequently, Bayesian maximum entropy (BME), a nonlinear modern spatiotemporal geostatistical model, is proposed to simultaneously consider the temporal and spatial autocorrelation of AOD for fusion. Compared with classical geostatistical methods, BME can integrate different types of data with uncertainty and significantly improve the estimation accuracy [40], [41]. Currently, BME has been widely used in soil [42], [43], hydrology [44], [45], [46], environment [47], [48], [49], and other scientific fields [50], [51].

Several studies have demonstrated that the introduction of BME results in a noticeable improvement in the spatial completeness of the fused AOD, and an acceptable accuracy can be obtained in regions where no valid observations are available [52], [53], [54]. However, numerous publicly released AOD products are available, and related validation efforts revealed significant accuracy differences and redundancy features among multiple products [18], [21], [26], which are not explicitly considered by BME, thus making the accuracy and continuity of the spatial distribution of fusion results still limited. Additionally, computational consumption increases significantly when larger amounts of AOD products are considered [35]. Our previous study constructed an accuracy ranking matrix of 16 publicly available products for different scenarios by comparing the accuracy under different surface types, aerosol loadings, and particle size types, which provides a priori knowledge about product accuracy for multisource product fusion [18]. Also, the study demonstrates that the fusion results based on this prior knowledge, i.e., the accuracy ranking matrix, can effectively combine the advantages of different products, which is a single-pixel-based fusion method. According to this conclusion, we conceive that instead of using all AOD products directly as BME inputs, the combined results of multiple AOD products based on the accuracy ranking matrix as BME inputs can effectively mitigate the influence of accuracy differences and redundancy characteristics among products. Also, the computational efficiency will be improved due to the reduction of input data.

Therefore, the main purpose of this article is to investigate whether the introduction of a priori knowledge about the accuracy of different products, i.e., the accuracy ranking matrix, can improve the quality of the BME estimation results compared to direct fusion using BME. The fusion experiment is conducted based on eight AOD products over China from different satellite sensors. Specifically, the overall accuracy of the two fusion results (with and without the accuracy ranking matrix) is compared based on ground-based observations in China. Meanwhile, we also analyze the applicability of the accuracy ranking matrix in different scenarios by comparing the accuracies of the two fusion results under different surface and atmospheric conditions. Furthermore, the improvement of the spatial distribution pattern of the fusion results by the introduction of the accuracy ranking matrix is discussed through case studies of the BTH and YRD regions.

II. Data

In this study, we used eight satellite AOD products, ground-based AOD measurements, and some auxiliary data over China from 2019 to 2021. The data from 2019 to 2020 were used for the determination of relevant parameters in the fusion process, and the data from 2021 were used for testing the effectiveness of the fusion method.

A. Satellite AOD Products

Our previous study demonstrated the complementary strengths of 16 AOD products from different satellite sensors over China [18]. Here, to ensure the temporal representativeness of the fusion data, eight of these AOD products that transit in the morning are selected for the fusion experiments. The eight products are from MODIS, MISR, Sentinel-3, and AHI sensors, and Table I provides a brief summary of these products.

The MODIS sensor onboard the Terra (approximately ~10:30 A.M. local time) satellite released five Level 2 AOD products based on three retrieval algorithms, namely the Multi-Angle Implementation of Atmospheric Correction (MAIAC), Dark Target (DT), and Deep Blue (DB) algorithms. Specifically, the MAIAC algorithm is based on time series analysis to obtain 1 km AOD products [55]. The DT algorithm has released two products with different resolutions, 3 and 10 km, both of which are retrieved for dark targets, with the difference being the choice of the retrieval window size [56], [57], [58]. Compared to the DT algorithm, the DB algorithm is initially designed to retrieve 10 km AOD products over bright surfaces [59], [60], and as the algorithm evolves, it can be extended to other surfaces such as vegetated areas [61]. Furthermore, the official 10 km merged DT and DB (DTB) product is also released by combining the advantages of

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TABLE I
SUMMARY OF SATELLITE AOD PRODUCTS USED IN THE FUSION EXPERIMENT

Sensor/platform	Product	SDS name	Quality control	Resolution	Version	Wavelength
MODIS/Terra	MOD04_3K	Optical_Depth_Land_And_Ocean	QA = 3	10:30, 3 km	C6.1	550 nm
	MOD04_L2 DT	Optical_Depth_Land_And_Ocean	QA = 3	10:30, 10 km	C6.1	550 nm
	MOD04_L2 DB	Deep_Blue_Aerosol_Optical_Depth_550_Land_Best_Estimate	QA ≥ 2	10:30, 10 km	C6.1	550 nm
	MOD04_L2 DTB	AOD_550_Dark_Target_Deep_Blue_Combined	_	10:30, 10 km	C6.1	550 nm
	MAIAC	Optical_Depth_055	highest quality	10:30, 1 km	C6	550 nm
MISR/Terra	MIL2ASAE	Aerosol_Optical_Depth	_	10:30, 4.4 km	V23	550 nm
Sentinel-3A/B	SYN L2 SDR	AOD_550	_	10:00, 4.5 km	1.0	550 nm
AHI/Himawari-8	L3	AOT_Merged	good and very good	Hourly, 5 km	V31	500 nm

DTB products based on the Normalized Difference Vegetation Index (NDVI) [12]. The latest version at 550 nm and quality-filtered products are used in this study, namely Collection 6.1 3 km DT, 10 km DT, DB, DTB, and Collection 6 MAIAC. The AOD product from the MISR sensor on board the Terra satellite is the Level 2 AOD product with a 4.4 km resolution, which is retrieved based on multiangle observation information [13]. The latest version (V23) of the MISR AOD product at 550 nm is selected for this study. Sentinel-3 (3A and 3B) transits at approximately 10:00 A.M. local time, which uses the overlapping observations of the carried OLCI and SLSTR sensors to retrieve the Level 2 Synergy aerosol products with a 4.5 km resolution [14], where 550 nm AOD products are selected. The AOD product from the AHI sensor on board the geostationary satellite Himawari-8 is the Level 3 hourly 5 km AOD_merge product, which has been processed to reduce the influence of cloud contamination and fill in missing values [15], [62]. The latest version (V31) and quality-screened AOD_merge product are selected. The AHI does not provide the 550 nm AOD. To be consistent with other products, the 550 nm AOD is obtained using the Ångström exponent (AE) and the 500 nm AOD_merge [63], [64], as follows:

$$\mathrm{AOD}_{550} = \mathrm{AOD}_{500}(550/500)^{-a} \tag{1}$$

where a is AHI AE_merged. Also, the values of 10:00 and 11:00 are averaged to match the transit times of other products. A detailed description of the satellite AOD products used in this study can be found in our previous study [18].

B. Ground-Based Data

In this study, the ground-based AOD measurements from the Aerosol Robotic Network (AERONET) and the National Civil Space Infrastructure Satellite Aerosol Product Validation Network (SIAVNET) are used for the construction of the accuracy ranking matrix, and the evaluation and analysis of the fusion results.

AERONET, a global aerosol ground-based observation network, provides accurate AOD measurements between 340 and 1640 nm approximately every 15 min with an uncertainty of approximately 0.01–0.02 [65]. The Level 1.5 AOD and AE measurements of Version 3 are used in this study. A total of 24 sites with valid observations from 2019 to 2021 are available in China (Fig. 1). SIAVNET

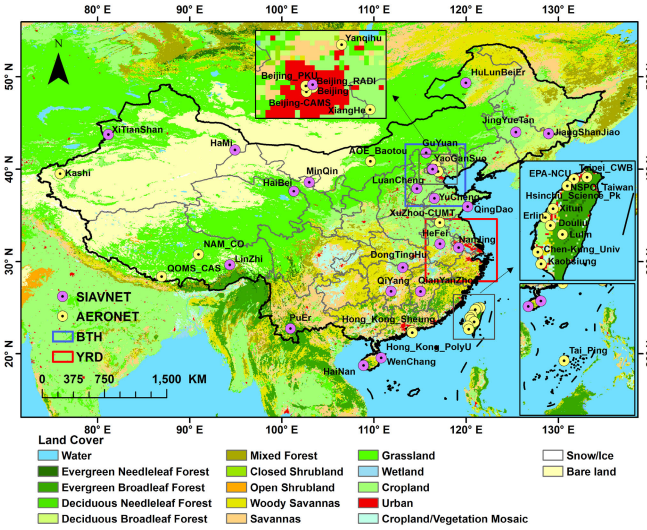


Fig. 1. Location of AERONET (yellow dots) and SIAVNET (purple dots) in China. BTH and YRD represent the BTH and YRD regions, respectively. Background map refers to the surface type obtained by the MODIS MCD12C1 product.

is the ground-based aerosol observation network in China organized by the Chinese Academy of Sciences, which effectively compensates for the limitations of sparse and uneven distribution of AERONET sites in China. It has been demonstrated that both have almost the same observation accuracy [66]. Here, SIAVNET Level 1.5 measurement data at 21 sites for 2019–2021 are used (Fig. 1).

Since AERONET and SIAVNET do not provide AOD observations at 550 nm, AOD observations at 550 nm are obtained using a robust quadratic polynomial fit based on valid AOD observations from 440 to 870 nm to be comparable with satellite AOD products [63], [67], as follows:

$$\log(\mathrm{AOD}_{550}) = a_0\log(550) + a_1\log(550)^2 + b \tag{2}$$

where a_0, a_1, and b are the fitting coefficients.

C. Auxiliary Data

Besides satellite and ground-based AOD products, several surface and aerosol parameter data are also needed in the accuracy ranking matrix construction and fusion process, including NDVI, digital elevation model (DEM), and AE data. The 16-day composited 1 km NDVI product from MODIS is

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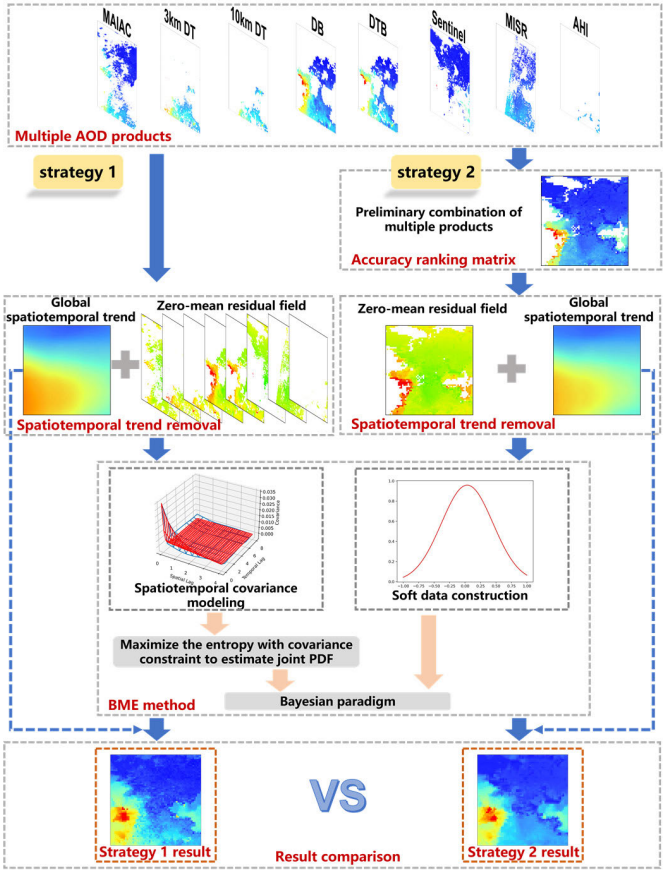


Fig. 2. Schematic illustration of two different multiple AOD product fusion strategies based on the BME approach.

selected to indicate surface cover conditions. The daily NDVI data are obtained based on the spline interpolation method, which has been widely used in previous studies [20], [68]. The DEM data with a 90 m resolution from the Shuttle Radar Topography Mission (SRTM) are used to reflect topographic information. The hourly aerosol AE data (470–870 nm) are from the Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2) with a spatial resolution of 0.625° (longitude) × 0.5° (latitude), which is used to represent different aerosol particle types.

III. Methodology

To investigate whether the introduction of an accuracy ranking matrix can improve the quality of BME estimation results, we fuse eight satellite AOD products in 2021 based on two fusion strategies, i.e., with and without the involvement of an accuracy ranking matrix, and conduct a comparative analysis of the two fusion results with a 10 km resolution. As illustrated by the flowchart in Fig. 2, the first fusion strategy is to directly incorporate the eight AOD products as the BME input (strategy 1). In contrast, the other is to initially combine the eight AOD products based on the accuracy ranking matrix, and then consider the combined results as the BME input (strategy 2). The main methods used in these two fusion strategies are described as follows.

A. Accuracy Ranking Matrix

The accuracy ranking matrix is based on a priori knowledge about the accuracy of multiple products and was originally constructed from our previous research using the same criteria to compare the accuracy of multiple satellite AOD products under different scenarios [18]. Also, it has been demonstrated that the fusion results based on it can effectively combine the advantages of different products. Compared to that work, in this study, more ground-based observations (SIAVNET) are considered and the construction method of the accuracy ranking matrix is further optimized. To evaluate the fusion results more reasonably, satellite AOD products and ground-based observations from 2019 to 2020 are used to reconstruct the accuracy ranking matrix, and data from 2021 are used for subsequent fusion experiments. Details of the construction of the accuracy ranking matrix are as follows.

1) To achieve uniform spatial resolution, eight satellite AOD products and auxiliary data, including NDVI, DEM, and AE data, are resampled to 10 km. For MERRA-2 AE data, the cubic convolution method is used to obtain 10 km data due to its coarse resolution [31]. While for all other data, the aggregation method is employed to maintain the original product accuracy as much as possible [20].
2) The eight satellite AOD products from 2019 to 2020 are validated using AERONET and SIAVNET observations. Here, the previously commonly used matching strategy is adopted, i.e., the satellite AOD values are averaged over 3 × 3 pixels centered at the ground-based sites, and the ground-based AOD values use the average of valid measurements from 10:00 A.M. to 11:00 A.M. local time [18], [20], [29]. Furthermore, the Global Climate Observing System (GCOS) fraction, which is the proportion of AOD retrievals that satisfy the climate research requirements, is used to represent the accuracy of AOD products. The higher the GCOS fraction of a product, the better it can meet the needs of climate research users [8], [67], [69]

$$\mathrm{GCOS} = \mathrm{maximum}\left(0.03, 0.1 \times \mathrm{AOD}_{\mathrm{ground}}\right). \quad (3)$$

3) Simultaneously considering the effects of surface and atmospheric conditions, the matched pairs are divided into different scenarios to obtain the accuracy (GCOS fraction) of all products in each scenario, i.e., the accuracy ranking matrix (Table II). Here, the scenario classification method is optimized to ensure a sufficient number of matched pairs in each scenario. Specifically, based on NDVI, surface cover is classified into three types: arid surface (NDVI $\leq$ 0.2), sparse-moderate vegetation cover (0.2 $<$ NDVI $<$ 0.6), and dense vegetation cover (NDVI $\geq$ 0.6) [20], [70]. Based on DEM, the surface elevation is divided into low elevation (DEM $\leq$ 800 m) and high elevation (DEM $>$ 800 m) [21], [71]. Based on the AOD and AE data provided by ground-based observations, the aerosol loading and types are classified into background conditions (AOD $\leq$ 0.2), coarse modal dominance

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TABLE II

ACCURACY (GCOS FRACTION) RANKING MATRIX DERIVED BASED ON SATELLITE AOD PRODUCTS FOR 2019–2020 OVER CHINA. PRODUCTS WITH THE HIGHEST ACCURACY IN EACH SCENARIO ARE SHOWN IN BOLD, AND "—" INDICATES THAT THE NUMBER OF MATCHING PAIRS IN THIS SCENARIO IS LESS THAN 5

NDVI	DEM	AOD/AE	MAIAC	3km DT	10km DT	DB	DTB	Sentinel	MISR	AHI
⩽0.2 arid surfaces	⩽800	background	38.1	12.1	0	23.1	23.1	36.0	**87.9**	13.2
		corse-dominated	**42.1**	28.6	20.0	20.0	17.1	0	28.6	30.3
		fine-dominated	**34.5**	11.1	—	7.3	8.4	10.0	18.8	25.6
	>800	background	31.5	15.1	0	30.7	30.7	40.6	**58.1**	0
		corse-dominated	8.9	9.5	0	7.3	4.9	**26.9**	22.2	18.2
		fine-dominated	**17.6**	—	—	15.4	15.4	0	—	0
0.2-0.6 sparse-moderate vegetation cover	⩽800	background	38.3	9.1	17.3	33.3	19.4	17.9	**58.0**	12.8
		corse-dominated	25.0	21.0	16.6	16.7	20.1	9.4	**27.0**	25.4
		fine-dominated	33.2	19.4	19.6	20.6	19.2	22.8	**42.4**	28.8
	>800	background	41.3	34.0	45.1	30.5	42.3	37.1	**53.8**	—
		corse-dominated	13.0	19.0	**41.2**	0	0.0	11.1	14.3	16.7
		fine-dominated	**40.7**	22.2	11.8	5.0	13.0	0	—	10.0
⩾0.6 dense vegetation cover	⩽800	background	26.4	22.3	35.3	28.4	35.7	3.6	**65.4**	0
		corse-dominated	15.2	**26.5**	16.0	7.4	16.0	0	—	23.1
		fine-dominated	**35.5**	29.6	32.3	16.3	31.8	15.1	20.7	24.7
	>800	background	10.1	18.5	25.9	**71.3**	25.9	3.9	65.2	0
		corse-dominated	—	—	—	—	—	—	—	—
		fine-dominated	37.5	28.1	29.6	**40.0**	29.6	—	—	39.1

(AOD > 0.2, AE ≤ 1), and fine modal dominance (AOD > 0.2, AE > 1) [67].

4) The eight satellite AOD products for 2021 are merged based on the accuracy ranking matrix (Table II) to combine the advantages of the different products. In brief, the product with the highest accuracy in each scenario is selected as the merged value, and if there is no valid value for this product, it is deferred downward. The scene that each pixel belongs to is determined by NDVI, DEM, MERRA-2 AE, and eight AOD product averages. In particular, the validation based on ground-based observations indicates that MERRA AE data perform well with a correlation coefficient of 0.522 and 76.8% of the values fall within the expected error (EE, ±0.04) envelope [72] (Fig. S1, Supplementary Material). In addition, it is worth noting that since the classification of scenarios is a range rather than a definite value, some uncertainty is acceptable for several products involved in the classification.

B. Spatiotemporal Trend Removal

The BME is based on the spatiotemporal random field theory for spatiotemporal analysis and mapping, which requires that the variables satisfy the assumption of second-order stationarity in the whole spatiotemporal domain [51]. Therefore, before investigating the autocorrelation structure of the data, it is necessary to quantify and remove the global trend to ensure that the data are regionalized random variables.

The AOD observation $X(p)$ at the specified spatiotemporal location p (s, t) can be decomposed into a global spatiotemporal trend $u(p)$ and a zero-mean residual field $\varepsilon(p)$ (2). $\varepsilon(p)$ represents the stochastic residual component, which can be used to describe the spatiotemporal autocorrelation of the variable

$$X(p) = u(p) + \varepsilon(p). \tag{4}$$

To remove the global trend, a spatiotemporal filtering window of size $49(x) \times 49(y) \times 3(t)$ is applied to the AOD data [52]. The global trend is obtained by calculating the average value of the AOD within the window, and then the stochastic residual component is obtained by subtracting the global spatiotemporal trend. In fusion strategy 1, the global spatiotemporal trend is obtained by applying the filter window to the mean values of the eight resampled AOD products and then generating the residual components of the eight products. While for fusion strategy 2, it is derived by applying the filter window to the fusion results based on the accuracy ranking matrix, which in turn yields the residual components corresponding to the fusion results.

C. BME Fusion Framework

The Bayesian method is a nonlinear estimator that integrates various types of physical knowledge and knowledge concepts, which can accept multiple spatiotemporal data with uncertainty under the theoretical framework. Given the existence of multiple proximity valid observations, the nonlinear mean estimate of the AOD residual component $\overline{X_p}$ for the spatiotemporal point p(s, t) can be obtained from the following equation [40]:

$$\overline{X_p} = \int X_p f\left(X_p \mid X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n}\right) \mathrm{d}X_p \tag{5}$$

where $X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n}$ are Gaussian probability soft data generated by satellite AOD products. The uncertainty in the soft data (i.e., variance) is determined by the comparison of satellite AOD products with ground-based observations from 2019 to 2020 [51], [53]. In fusion strategy 1, these soft data are directly provided by the eight detrended satellite AOD products. In fusion strategy 2, it is provided by the detrended merged results based on the accuracy ranking matrix. $f(X_p|X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n})$ represents the posterior

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probability density function (pdf) at location p, under the adjacent valid observations.

Under the Bayesian rule, the posterior pdf at the estimation point is updated from the prior pdf when soft data are considered, expressed as follows:

$$\begin{aligned} & f\left(X_p \mid X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n}\right) \\ &= \frac{f\left(X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n}, X_p\right)}{f\left(X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n}\right)} \\ &= \frac{f_G\left(X_{\text{map}}\right)}{f\left(X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n}\right)} \end{aligned} \tag{6}$$

where $\left(X_{\text{soft1}}, X_{\text{soft2}}, \ldots, X_{\text{soft}n}\right)$ is the a priori pdf of the residual components of adjacent AOD observations. $f_G(X_{\text{map}})$ is the joint pdf obtained by maximizing the information entropy under the general knowledge constraint [40]. In this study, general knowledge G is the spatiotemporal correlation of the satellite AOD residual components, described by two nested exponential spatiotemporal covariance models [52], [53]

$$\begin{aligned} G(d, \tau) = & c_1 \exp\left(-\frac{3d}{a_{s1}}\right) \exp\left(-\frac{3\tau}{a_{t1}}\right) \\ & + c_2 \exp\left(-\frac{3d}{a_{s2}}\right) \exp\left(-\frac{3\tau}{a_{t2}}\right). \end{aligned} \tag{7}$$

Here, d and τ represent the spatial and temporal lags between different estimation points. c_1 and c_2 are the partial basis variances of the two exponential models, respectively, representing the contribution of the models to the total variance of the process. a_{s1} and a_{s2} are the spatial ranges of the two exponential models, and a_{t1} and a_{t2} correspond to the temporal ranges of the two exponential models. Here, the range is a distance parameter, and when the distance (d) or time (τ) is greater than the range, the spatial or temporal correlation disappears. The free parameters in the model, i.e., c and a, are obtained by fitting the AOD residual components with the least squares method. The spatial and temporal ranges in the exponential model with larger c are used as the spatiotemporal search distance threshold in the subsequent BME fusion process. In fusion strategy 1, the spatiotemporal covariance model is fit using the mean values of the eight detrended AOD products, while in strategy 2, it is obtained by the detrended merged results based on the accuracy ranking matrix.

For continuous variables, the information entropy H is defined as follows:

$$H = -\int dX_{\text{map}} f_G\left(X_{\text{map}}\right) \log f_G\left(X_{\text{map}}\right). \tag{8}$$

The Lagrange function is constructed by introducing a Lagrange multiplier μ. Maximizing the information entropy H is equivalent to maximizing the Lagrange function

$$\begin{aligned} & L\left[f_G\left(X_{\text{map}}\right)\right] \\ &= -\int dX_{\text{map}} f_G\left(X_{\text{map}}\right) \log f_G\left(X_{\text{map}}\right) \\ & \quad - \sum_{\alpha=1}^{N} \mu_\alpha \left[\int G_\alpha\left(X_{\text{map}}\right) f_G\left(X_{\text{map}}\right) dX_{\text{map}} - \overline{G_\alpha\left(X_{\text{map}}\right)}\right] \end{aligned} \tag{9}$$

where $\overline{G_\alpha(X_{\text{map}})}$ is the expected value of $G_\alpha(X_{\text{map}})$. The maximum entropy solution of the joint pdf can be derived by setting the partial derivatives to zero

$$f_G\left(X_{\text{map}}\right) = \frac{\exp\left(\sum_{\alpha=1}^{N} \mu_\alpha G_\alpha\left(X_{\text{map}}\right)\right)}{\int \exp\left(\sum_{\alpha=1}^{N} \mu_\alpha G_\alpha\left(X_{\text{map}}\right)\right) dX_{\text{map}}}. \tag{10}$$

Therefore, the AOD residual component $\overline{X_p}$ at the spatiotemporal estimation point p can be solved by combining (5)–(10). Also, the AOD value is obtained by adding the corresponding global trend.

D. Fusion Accuracy Evaluation

In this study, we used two fusion strategies to fuse eight satellite AOD products of 2021 over China and evaluated the two fusion results with ground-based observation data from AERONET and SIAVNET. Here, the validation methodology used is consistent with that used for the eight original satellite AOD products in the accuracy ranking matrix construction process. In addition to the GCOS fraction, the EE fraction designed for the MODIS DT algorithm [12], correlation coefficient (R), mean absolute error (MAE), root mean square error (RMSE), and relative mean bias (RMB) were also used to describe the performance of the fusion results

$$\text{EE_DT} = \pm\left(0.05 + 0.15 \times \text{AOD}_{\text{ground}}\right) \tag{11}$$

$$\text{RMB} = \frac{1}{N} \sum_{i=1}^{N} \left(\text{AOD}_{\text{merge}} / \text{AOD}_{\text{ground}}\right). \tag{12}$$

Since the purpose here is to validate the accuracy of the two fusion results, we only obtained the estimated values at the site locations to compare with the ground-based AOD for faster computational efficiency. Specifically, for a site, we first create a spatial window of 4° × 4° size centered on that site. Then, a spatiotemporal covariance model is constructed based on the detrended satellite AOD data for 2021 within this window. Finally, for a given day, the BME method is used to obtain the estimated values of a total of 9 pixels in the 3 × 3 grid centered on the site, and the mean value of the 9 pixels is compared with the ground-based AOD observation. Here, the valid spatial range of the spatiotemporal covariance model is generally about 2° as demonstrated by existing studies [53] and numerous experiments in this study; thus, the spatial window of 4° × 4° can be satisfied with sufficient soft data for spatiotemporal covariance modeling.

IV. Experimental Results

A. Overall Accuracy Comparison

Using the ground-based measurements from AERONET and SIAVNET of China in 2021 as a reference, we obtained the accuracy and consumption time of the BME fusion results for both strategies (Fig. 3). The results suggest that compared with fusion strategy 1, the fusion accuracy is significantly improved after incorporating the accuracy ranking matrix, with the fraction of retrievals satisfying the GCOS requirement increasing by 12.2% (27.9%–31.3%) and the proportion falling within the EE envelope increasing by 6.5% (60.7%–64.9%).

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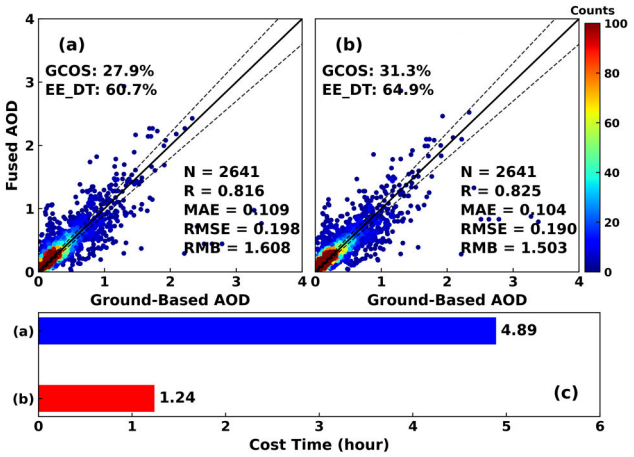


Fig. 3. Comparison of accuracy and computation time of two fusion strategies: (a) accuracy of fusion strategy 1; (b) accuracy of fusion strategy 2; and (c) consumption time of the two fusion strategies. The dashed line in the scatter plot is the error range defined by the GCOS requirement, and the solid black line is the 1:1 line.

The other validation metrics (i.e., R, MAE, RMSE, RMB) also show improvement, with RMB decreasing from 1.6 to 1.5, indicating that strategy 2 mitigates the overestimation of the fusion results. Furthermore, the accuracy comparison of strategy 1 (Fig. S2, Supplementary Material) and strategy 2 (Fig. S3, Supplementary Material) with the eight original products was also performed. Strategy 1 performs slightly worse than the MAIAC and MISR products, which are generally ranked higher in the accuracy ranking matrix, mainly due to the fact that BME does not explicitly consider the accuracy differences and redundancy features between products. In contrast, strategy 2 outperforms the MAIAC product and is comparable to the MISR, indicating that it adequately combines the strengths of the different products. In terms of the computation time for this experiment, the introduction of the accuracy ranking matrix significantly improves the running efficiency under the same computer configuration, reducing the running time from 4.89 to 1.24 h, with an efficiency improvement of 74.6% [Fig. 3(c)]. These results show that fusion strategy 2 not only improves the computational efficiency by reducing the input data but also enhances the estimation accuracy by explicitly combining the advantages of different products.

B. Accuracy Comparison Under Different Scenarios

In this section, to analyze the applicability of the accuracy ranking matrix in different scenarios, we compare the accuracy of the two fusion strategies under different surface conditions and different atmospheric conditions.

1) Different Surface Conditions: Since NDVI is an indicator of the surface conditions, here, we use NDVI to classify the surface cover types [69]. To ensure sufficient matching pairs in each type, the surface cover is simplified into two types based on NDVI: arid surface cover (NDVI $\leq$ 0.2) and vegetated surface cover (NDVI $>$ 0.2) [20], [69]. Fig. 4(a) shows the bias distribution of the two fusion strategies for different surface conditions. Compared with strategy 1, the

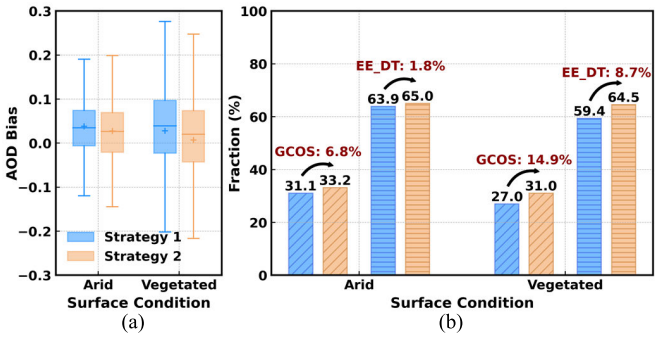


Fig. 4. Accuracy comparison of the two fusion strategies under different surface conditions. (Left) Box plots of AOD bias for the two fusion strategies under different surface conditions. In each box, the middle, upper, and lower horizontal lines represent the median, 25th, and 75th percentile of AOD bias, respectively. The plus sign indicates the AOD bias mean value. (Right) GCOS fraction (diagonal lines filled) and EE_DT fraction (horizontal lines filled) for the two fusion strategies under different surface conditions.

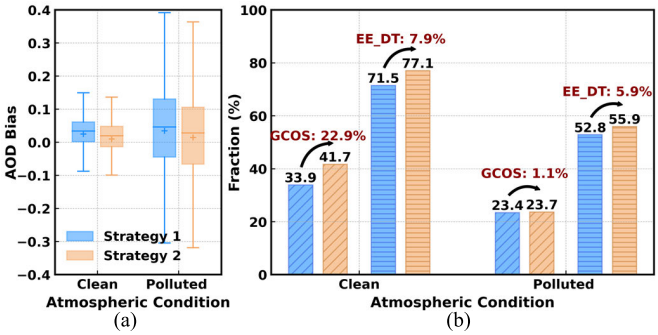


Fig. 5. Accuracy comparison of the two fusion strategies under different atmospheric conditions. (Left) Box plots of AOD bias for the two fusion strategies under different atmospheric conditions. In each box, the middle, upper, and lower horizontal lines represent the median, 25th, and 75th percentile of AOD bias, respectively. The plus sign indicates the AOD bias mean value. (Right) GCOS fraction (diagonal lines filled) and EE_DT fraction (horizontal lines filled) for the two fusion strategies under different atmospheric conditions.

overestimation of BME estimation results under both surface covers is alleviated by introducing the accuracy ranking matrix, and the alleviation is more obvious in vegetated areas, with the bias mean value closer to zero. Additionally, for arid and vegetation cover, 33.2% and 31% of the fusion strategy 2 satisfy the GCOS requirements as well as 65% and 64.5% fall within the EE_DT envelope, respectively [Fig. 4(b)]. Both of these indicators indicate that the fusion accuracy of strategy 2 is higher than that of strategy 1 (GCOS fraction: 31.1%, 27%; EE_DT fraction 63.9%, 59.4%). The comparison of accuracy differences between the two strategies under different surface conditions shows that the introduction of the accuracy ranking matrix in vegetation cover areas results in more significant accuracy improvement, where the GCOS fraction is increased by 14.9% and the EE_DT fraction is increased by 8.7%. Therefore, for different surface conditions, the accuracy ranking matrix has better applicability in vegetated areas than in arid areas. The poor applicability under arid surfaces may be because some algorithms, such as DT, fail to retrieve on bright surfaces [12], [57], leading to fewer products with valid values in such scenarios, thus making the accuracy ranking matrix less useful.

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TABLE III
SPATIAL COVERAGE RATIO OF THE EIGHT ORIGINAL PRODUCTS AND TWO FUSION RESULTS CORRESPONDING TO THE CASES OF BTH AND YRD REGIONS

	BTH		YRD	
	02/12/2021 (%)	04/13/2021 (%)	04/17/2021 (%)	04/19/2021 (%)
MAIAC	74.2	69.6	50.6	47.9
3km DT	41.0	38.3	63.1	49.7
10km DT	22.3	21.5	43.1	38.7
DB	45.8	43.3	47.8	36.1
DTB	49.4	37.0	43.1	38.7
Sentinel	9.8	67.5	60.4	59.0
MISR	18.3	6.4	21.0	31.0
AHI	8.8	0.4	23.5	26.4
Strategy 1	100	100	100	100
Strategy 2	100	100	100	100

2) Different Atmospheric Conditions: In order to discuss the applicability of the accuracy ranking matrix under different atmospheric conditions, we divided the atmospheric conditions into two types: clean air conditions (AOD $\leq$ 0.2) and polluted conditions (AOD $>$ 0.2) [67]. The accuracy comparison of the two fusion strategies under different atmospheric conditions is shown in Fig. 5. Whether in clean or contaminated conditions, the bias of the fusion results is decreased and the overestimation phenomenon is mitigated by the introduction of the accuracy ranking matrix [Fig. 5(a)]. In terms of the GCOS and EE_DT fraction, with the introduction of the accuracy ranking matrix under the polluted conditions, the proportion of fusion results satisfying the GCOS and EE_DT requirements improved by 1.1% (23.4%–23.7%) and 5.9% (52.8%–55.9%), respectively [Fig. 5(b)]. However, in contrast, the accuracy improvement is more evident in clean conditions, where the GCOS fraction is increased by 22.9% (33.9%–41.7%) and the EE_DT fraction is increased by 7.9% (71.5%–77.1%) [Fig. 5(b)], which indicates that the accuracy ranking matrix is significantly more appropriate for clean conditions than for polluted conditions. The noticeable accuracy improvement in clean atmosphere conditions may be attributed to the significant differences in the performance of different AOD products under this condition [18].

C. Spatial Distribution Differences in Typical Regions

A major strength of the BME method is the ability to obtain AOD distributions with high spatial completeness based on spatiotemporal autocorrelation, which is important for aerosol variability analysis and climate-related studies [53]. To further compare the spatial patterns of the fusion results derived from the two strategies, we selected the BTH and YRD regions, which are characterized by high population density and high aerosol loading [73], [74], for the case study. Based on different fusion strategies, two types of daily AOD distributions in 2021 were obtained over BTH and YRD regions. Here, for both BTH and YRD, two typical cases are selected for spatial comparison, i.e., the polluted scene and the relatively clean scene.

Fig. 6 illustrates the comparison of the spatial distribution of the two fusion strategies in BTH for the high aerosol loading scenario on 12 February 2021 [Fig. 6(a) and (b)] and the relatively clean scenario on 13 April 2021 [Fig. 6(c) and (d)]. In both cases, the fused AODs derived from two strategies could achieve a spatial coverage rate of 100%. In contrast, data gaps existed in all eight original AOD products, with completeness rates of 8.8%–74.2% on 12 February and 0.4%–69.6% on 13 April (Table III, Figs. S4 and S5, Supplementary Material). Although the two fusion strategies show a similar pattern on the whole, obvious differences still exist. As seen in Fig. 6(a) and (c), the fusion results of strategy 1 show patchy patterns and unsmooth spatial variation, which may be due to the large differences among products [18]. However, with the introduction of the accuracy ranking matrix, the patchy phenomenon of the fusion results is alleviated, and the spatial variation tends to be smoother [Fig. 6(b) and (d)] and more similar to that of the original product (Figs. S4 and S5). In addition, Table IV summarizes the comparison of the fusion results with ground-based observations in two cases of BTH. Here, the mean value of 3 $\times$ 3 pixels centered on the site is taken as the fusion result value. Compared with strategy 1, both in the polluted case and in the relatively clean case, the fusion results of strategy 2 attenuate the overestimation phenomenon and are closer to the true values of the sites, indicating that the introduction of the accuracy ranking matrix can yield more reasonable fusion results.

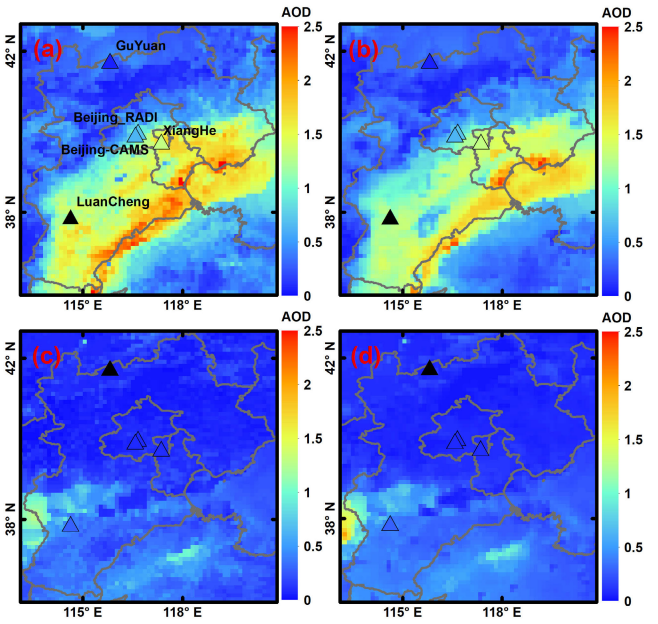


Fig. 6. Spatial distribution comparison of the fusion results obtained by two strategies over the BTH region on 12 February 2021 (polluted case, a, b) and 13 April 2021 (clean case, c, d). (a) and (c) Results obtained by fusion strategy 1. (b) and (d) Results derived from fusion strategy 2. The black triangle indicates no data.

The comparison of the spatial patterns of the two fusion results in YRD for the polluted scene on 17 April 2021, and the clean scene on 19 April 2021, is presented in Fig. 7. For both cases, the spatial gaps of the eight original AOD products are apparent, with spatial completeness of 21%–63.1% and 26.4%–59%, respectively (Table III, Figs. S6 and S7, Supplementary Material). However, the coverage rates of the fusion results obtained based on both strategies reached

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TABLE IV
COMPARISON OF THE GROUND-BASED MEASUREMENTS WITH THE TWO FUSION RESULTS FOR TWO CASES OVER THE BTH AND YRD REGIONS ("—" MEANS NO DATA)

	Polluted case			Clean case		
	Strategy 1	Strategy 2	Site Value	Strategy 1	Strategy 2	Site Value
Beijing_RADI	0.837	0.762	0.706	0.099	0.091	0.086
Beijing-CAMS	0.890	0.874	0.628	0.093	0.077	0.083
XiangHe	1.667	1.303	1.301	0.119	0.113	0.095
GuYuan	0.395	0.300	0.052	0.116	0.094	—
LuanCheng	1.246	1.186	—	0.501	0.451	0.285
HeFei	0.519	0.530	0.555	0.343	0.323	0.298
NanJing	0.774	0.797	0.808	0.352	0.333	0.274

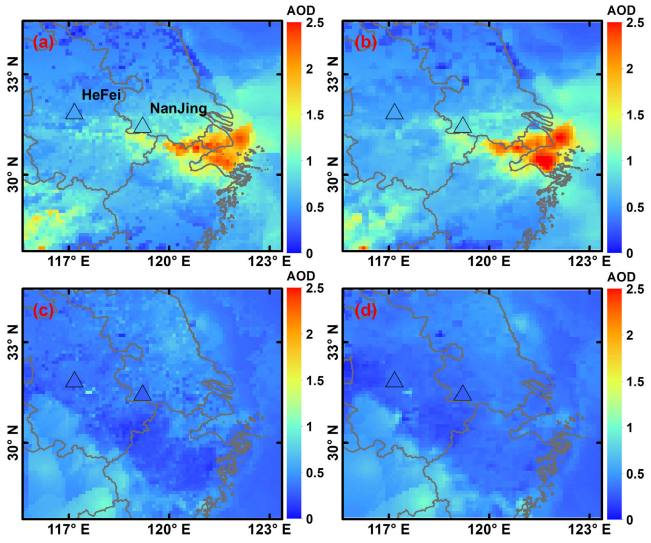


Fig. 7. Spatial distribution comparison of the fusion results obtained by two strategies over the YRD region on 17 April 2021 (polluted case, a, b) and 19 April 2021 (clean case, c, d). (a) and (c) Results obtained by fusion strategy 1. (b) and (d) Results derived from fusion strategy 2.

100% (Fig. 7). Similar to the case of BTH, the AOD spatial distribution in YRD obtained based on strategy 2 also weakens the patchy phenomenon in that obtained based on strategy 1, with a more continuous distribution (Fig. 7). Meanwhile, by comparing with the ground-based observations provided by two sites, Hefei and Nanjing, it can be seen that the results obtained by strategy 2 are more reasonable as they tend to be the true values (Table IV).

Therefore, the case studies of BTH and YRD can demonstrate that in the process of BME fusion, the introduction of the accuracy ranking matrix can lead to more realistic fusion results with smoother spatial variation than the direct fusion of multiple products.

V. CONCLUSION

The BME method can fuse multiple satellite AOD products to generate results with high spatiotemporal coverage based on the spatiotemporal autocorrelation of the data, which allows for better application to climate-related studies. However, the large accuracy differences and redundant characteristics among satellite AOD products still limit the quality of BME fusion results. Given that the accuracy ranking matrix, as the prior knowledge about product accuracy, explicitly describes the accuracy comparison of different AOD products in different scenarios, we try to introduce it in BME fusion to fully combine the advantages of different products and obtain higher quality fusion results. Specifically, in this study, the fusion experiment is conducted based on eight satellite AOD products over China. The results of two fusion strategies are compared, one is the direct fusion of multiple products using BME, and the other is the introduction of the accuracy ranking matrix.

The comparison of the fusion results from the two strategies with the ground-based observations from AERONET and SIAVNET demonstrates that the introduction of the accuracy ranking matrix yields more accurate results. Compared to directly fusing, the proportion of results that satisfied the GCOS and EE requirements increases from 27.9% and 60.7% to 31.3% and 64.9%, respectively, with an improvement of 12.2% and 6.5%, respectively. Other accuracy metrics, i.e., R, MAE, RMSE, and RMB, also improve, proving that the introduction of the accuracy ranking matrix can significantly improve the fusion accuracy. In addition, the computational efficiency is also significantly promoted due to the reduced number of input data for BME.

By analyzing the accuracy differences between the two fusion results under different surface conditions and different atmospheric conditions, it can be found that in all scenarios, the accuracy of the fusion results is all improved after introducing the accuracy ranking matrix. In particular, the accuracy improvement in vegetated regions and clean air conditions is higher than that in arid regions and polluted conditions, with the GCOS fraction improving by 14.9%, 22.9%, 6.8%, and 1.1%, respectively, and the EE_DT fraction improving by 8.7%, 7.9%, 1.8%, and 5.9%, respectively. These results indicate better applicability of the accuracy ranking matrix in vegetated areas and clean atmospheric conditions.

To further analyze the impact on the spatial pattern of fusion results by introducing the accuracy ranking matrix, the differences in the spatial distribution of the two strategies are discussed based on the cases of polluted and clean days in the BTH and YRD regions. The results show that the introduction of the accuracy ranking matrix can effectively alleviate the patchy phenomenon in the direct fusion results and make the spatial variation smoother. Moreover, the comparison with the ground-based observations in the region shows that the fusion results with the introduction of the accuracy ranking matrix are more consistent with reality.

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Overall, for BME-based fusion of multiple satellite AOD products, the introduction of the accuracy ranking matrix, i.e., a priori knowledge about the accuracy of different products, can effectively mitigate the effects of accuracy differences and redundant features among products by explicitly combining the advantages of different products to obtain higher quality fusion results with less time consumption.

Currently, numerous aerosol products with complementary strengths are publicly released. The BME method can integrate different types of data with uncertainty. In this study, only several aerosol products from passive satellite sensors were used. Besides, active sensors, such as Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP), can provide aerosol data below clouds [52]. Reanalysis datasets, such as MERRA-2, provide complete coverage of aerosol distributions, despite large uncertainties [75]. Ground-based observations, although sparsely distributed, can provide accurate aerosol information. The reconstruction accuracy of large areas with missing data can be effectively improved by fully utilizing these data. Therefore, various types of data need to be efficiently fused to generate a harmonized and high-quality aerosol dataset to support climate-related studies, which can be well achieved by the method proposed in this study. This is the focus of our forthcoming work. Meanwhile, we will further optimize the fusion method and investigate other possible ways of introducing a priori knowledge in BME fusion. In addition, since the BME method is also widely used in other remote-sensing product fusion studies, the findings of this study are expected to provide novel ideas for these studies.

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